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Transport Modeling of Scale Inhibitor and Scaling Ions in an Oil-Water Trunkline with Multiple Flowlines

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Abstract

Operators transport produced oil-water mixtures through trunklines fed by multiple production flowlines, a step that is critical to every upstream operation. As operating conditions shift, inorganic scales readily precipitate and deposit on the inner walls of both flowlines and trunklines as the concentration of different scaling ions increases. The inhibitor's effectiveness depends on its local concentration along the pipe, yet routine sampling captures concentrations only at a few accessible points, typically at the trunkline outlet. We present a new CFD-based mass-transport model that simulates the concurrent transport of a water-soluble scale inhibitor and scaling ions in a trunkline fed by multiple flowlines. We resolve the coupled two-phase flow and multi-species transport of inhibitor plus scaling ions in a representative section of trunkline. Because the species equations couple tightly with the two-phase flow field through the velocity and turbulence terms, we solve the entire system of equations simultaneously. The concentration front transports downstream as time advances, showing a localized high-concentration plume near the injection line that dilutes downstream. Steady state is reached after approximately 1 h of injection under the specified operating conditions. Both inhibitor and ion curves exhibit discrete jumps (dilution for inhibitor, enrichment for scaling ions) where the incoming flowlines merge with the trunkline. Between tie-ins, turbulent mixing quickly flattens the profiles, confirming that turbulent mixing dominates over molecular diffusion for species transport. Including individual ions, rather than a lumped salinity term, captures local concentration spikes that drive precipitation thermodynamics. Injection concentration and flow rate therefore provide two independent operational parameters. This study demonstrates that the presented model and the CFD-based simulations enable the optimization of the scale inhibitor injection strategy, which matches site-specific water chemistry, tie-in geometry, and flow conditions.

Introduction

Operators transport produced oil-water mixtures through trunklines fed by multiple production flowlines, a step that is critical to every upstream operation (Su et al., 2022; Höhne et al., 2020; Ismail et al., 2015). As operating conditions shift, inorganic scales readily precipitate and deposit on the inner walls of both flowlines and trunklines as the concentration of different scaling ions increases (Liu et al., 2025). These deposits block flow, diminish hydraulic efficiency, and trigger equipment failures.

To combat deposition, field engineers inject water-soluble scale inhibitors continuously at the trunkline header. The inhibitor's effectiveness depends on its local concentration along the pipe, yet routine sampling captures concentrations only at a few accessible points, typically at the trunkline outlet. Without a full concentration profile, operators cannot optimize dose rates or injection locations.

Predicting that profile poses a complex multiphase-flow problem (Chen et al., 2023; Pouraria et al., 2021). The trunkline carries turbulent, oil-water two-phase flow whose velocity field, phase distribution, and turbulent mixing control how the inhibitor and dissolved ions spread and dilute. Advanced CFD tools therefore provide the only practical route to quantify inhibitor transport under realistic operating conditions.

Standard CFD frameworks couple the Navier–Stokes equations with species-transport equations, representing advection, diffusion, and turbulent mixing (Desamala et al., 2014; Dewangan et al., 2020). Researchers model the two immiscible phases with Eulerian–Eulerian, Volume-of-Fluid VOF, Level-Set, or Eulerian–Lagrangian approaches, while they resolve turbulence with either Reynolds-Averaged Navier–Stokes RANS closures or the more detailed Large Eddy Simulation LES (Li et al., 2021; Sunday et al., 2022). Prior studies (e.g., Su et al., 2022; Höhne et al., 2020; De la Cruz-Ávila et al., 2022; Nouri et al., 2021; Zhang et al., 2019) characterize velocity, pressure, and phase distribution, but rarely track inhibitor species together with a full suite of scaling ions. Moreover, most published models examine a single, uniform inflow, sidestepping the mixing that occurs when several flowlines discharge at different axial positions. Inter-line mixing radically alters local water chemistry and, by extension, inhibitor demand; yet literature offers little guidance (Torres-Monzón, 2005).

We present a new CFD-based mass-transport model that simulates the concurrent transport of a water-soluble scale inhibitor and scaling ions in a trunkline fed by multiple flowlines. The study quantifies how flow rate, and the number and placement of inlets (i.e., flowlines) control concentration field of the inhibitor and scaling ions. By resolving the turbulent oil-water flow and the coupled species transport, we supply practical insight for optimizing injection strategy and safeguarding pipeline throughput.

Mathematical Implementation

The governing equations presented below originate from our earlier publication (Liu et al., 2025) on single-inhibitor transport and are reproduced here for completeness; we now extend the framework to track scaling ions concurrently with the inhibitor and modify some equations for accuracy of simulation.

Flow Description: Two-Phase Mixture Model

We represent the trunkline and its connected flowlines as an isothermal, low-pressure system in which both oil and water densities remain constant. Using the mixture model, we treat the oil phase as continuous and the water phase as dispersed, while both phases share a single pressure field.

Continuity. The mixture density ρ combines the contributions of the continuous and dispersed phase via their volume fractions ϕ_c and ϕ_d :

$$\rho = \phi_c \rho_c + \phi_d \rho_d \quad (1)$$

$$\phi_d + \phi_c = 1 \quad (2)$$

where ρ_c and ρ_d are the densities of the continuous phase (oil) and dispersed phase (water). We define the volume flux and obtain the mixture-averaged velocity by adding the individual phase velocities:

$$\text{Volume flux: } j_d = \phi_d u_d \quad (3)$$

$$j_c = \phi_c u_c \quad (4)$$

$$\text{Volume-averaged velocity: } j = j_d + j_c \quad (5)$$

where u_c and u_d are the continuous and dispersed phase velocity vectors. Substituting these relations into the differential mass balance yields the mixture continuity equation:

$$\frac{\partial}{\partial t}(\phi_c \rho_c + \phi_d \rho_d) + \nabla \cdot (\rho_d \phi_d u_d + \rho_c \phi_c u_c) = 0 \quad (6)$$

Then, assuming constant phase densities Eq. (6) simplifies to the alternative form:

$$\nabla \cdot j = m_{dc} \left(\frac{\rho_c - \rho_d}{\rho_c \rho_d} \right) \quad (7)$$

where m_{dc} denotes mass transfer rate from the dispersed to the continuous phase.

Momentum. We define the momentum balance for the mixture:

$$\rho \frac{\partial}{\partial t}(j) + \rho(j \cdot \nabla)j + \rho_c r_d (j_{slip} \cdot \nabla)j = -\nabla p + \nabla \cdot \tau - \nabla \cdot K_m + F_m + \rho g + F \quad (8)$$

$$\tau = (\mu + \mu_T)(\nabla j + \nabla j^T) - \frac{2}{3}(\mu + \mu_T)(\nabla \cdot j)I - \frac{2}{3}\rho k I \quad (9)$$

$$K_m = \rho_c (1 + \phi_c r_d) u_{slip} j_{slip}^T \quad (10)$$

$$F_m = -\rho_c r_d \left[(j \cdot \nabla)j_{slip} + (\nabla \cdot (D_{md} \nabla \phi_d))j + j_{slip} m_{dc} \left(\frac{\rho_c - \rho_d}{\rho_c \rho_d} \right) \right] \quad (11)$$

$$j_{slip} = \phi_c \phi_d u_{slip} \quad (12)$$

$$\mu = \mu_d \phi_d + \mu_c \phi_c \quad (13)$$

where p is the common pressure shared by the continuous and dispersed phases, g is the gravitational accelerations, I is a unit vector, u_{slip} is the slip velocity vector between phases, j_{slip} is the slip flux, r is the sum of the stress tensors (viscous and turbulent), μ is the fluid viscosity, k is the turbulent kinetic energy, F is any additional volume force, and D_{md} is the turbulent dispersion coefficient. Lastly, the reduced density difference r_d is:

$$r_d = \frac{\rho_c - \rho_d}{\rho_c} \quad (14)$$

Turbulent Flow. Because the Reynolds number exceeds the turbulent transition threshold, we employ the standard k- ϵ model to simulate turbulence:

$$\text{Transport, } k\text{-equation: } \rho \frac{\partial k}{\partial t} + \rho(j \cdot \nabla)k = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_k} \right) \nabla k \right] + G_k - \rho \epsilon \quad (15)$$

$$\text{Transport, } \epsilon\text{-equation: } \rho \frac{\partial \epsilon}{\partial t} + \rho(j \cdot \nabla)\epsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\epsilon} \right) \nabla \epsilon \right] + C_{1\epsilon} \frac{\mu_T}{k} (G_k) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (16)$$

$$\text{Turbulent kinetic energy: } G_k = \mu_T \left[\nabla j : (\nabla j + (\nabla j)^T) - \frac{2}{3}(\nabla \cdot j)^2 \right] - \frac{2}{3}\rho k \nabla \cdot j \quad (17)$$

$$\text{Turbulent viscosity: } \mu_T = C_\mu \rho \frac{k^2}{\epsilon} \quad (18)$$

where G_b is the turbulent kinetic energy generated by buoyancy (equal to zero as both the continuous and dispersed phases are incompressible), σ_k and σ_ϵ are the Prandtl number of turbulent kinetic energy and turbulent energy dissipation rate, and $\sigma_k=1.0$, $\sigma_\epsilon=1.3$, $C_{1\epsilon}=1.44$, $C_{2\epsilon}=1.92$, $C_\mu=0.09$.

Phase Transport. We solve the convection–diffusion equation for the dispersed-phase volume fraction:

$$\frac{\partial}{\partial t}(\phi_d \rho_d) + \nabla \cdot (\rho_d \phi_d \mathbf{u}_d) = \nabla \cdot (\rho_d D_{md} \nabla \phi_d) - m_{dc} \quad (19)$$

Turbulent dispersion, parameterized by $D_{md} = \mu_T / \rho \sigma_T$ with Schmidt number $\sigma_T \approx 0.35 - 0.7$, augments molecular diffusion. Assuming constant dispersed-phase density, we rewrite the equation in conservative form:

$$\frac{\partial}{\partial t}(\phi_d) + j \cdot \nabla(\phi_d) + \nabla \cdot j_{slip} = \nabla \cdot (D_{md} \nabla \phi_d) - \frac{m_{dc} \rho}{\rho_d \rho_c} \quad (20)$$

with the continuous-phase fraction calculated using Eq. (2).

Mass Transport of Inhibitor and Scaling Ions. For modeling of mass transport of the scale inhibitor and scaling ions, we assume the following:

1. Water-soluble species only: inhibitor and all scaling ions reside exclusively in the water phase.
2. No partitioning or adsorption: we neglect wall adsorption or phase transfer.
3. No reaction coupling: we ignore inhibitor-ion interaction and precipitation kinetics.

Based on the assumption, we use convection–diffusion equation to describe the mass transport of the scale inhibitor and scaling ions:

$$\frac{\partial}{\partial t}(\phi_w C_k) + \nabla \cdot (-D_i \nabla(\phi_w C_k) + u_w \phi_w C_k) + \nabla \cdot j_{slip} = R_{inj,k} (k = 1, 2, \dots, n) \quad (21)$$

$$R_{inj,k} = u_{inj} \phi_w C_{inj,k} \quad (22)$$

where C is the concentration of the scale inhibitor or one of the scaling ions, u_{inj} and $C_{inj,k}$ are the injection velocity and concentration, and D is the diffusion coefficient of the scale inhibitor or the scaling ions, each species diffuses with a small coefficient D_i , so turbulent convection dominates transport

Because the species equations couple tightly with the two-phase flow field through the velocity and turbulence terms, we solve the entire system of equations simultaneously.

We specify all inlet volume fractions, velocities, and species concentrations for each inflow line, and set a uniform pressure at the trunkline.

Numerical Simulation

Geometry and Model Set Up

Figure 1 shows a schematic diagram of the modeled trunkline system. We use COMSOL Multiphysics 6.2 to resolve the coupled oil-water two-phase flow and multi-species transport of inhibitor plus two scaling ions in a representative section of trunkline (Fig. 1). The model consists of a horizontal, 1,000 meters long trunkline with three input sources, namely flowlines. The three production flowlines connect to the trunkline in a 12'o clock direction, and their dimensions are listed in Table 1.

Table 1—Trunkline system dimensions

Dimension	Trunkline	Inhibitor Injection Line	Flowline 1	Flowline 2	Flowline 3
Inner diameter (ID) [inch]	24	3/8	8	8	8
Length or distance [m]	1000	5	150	450	750

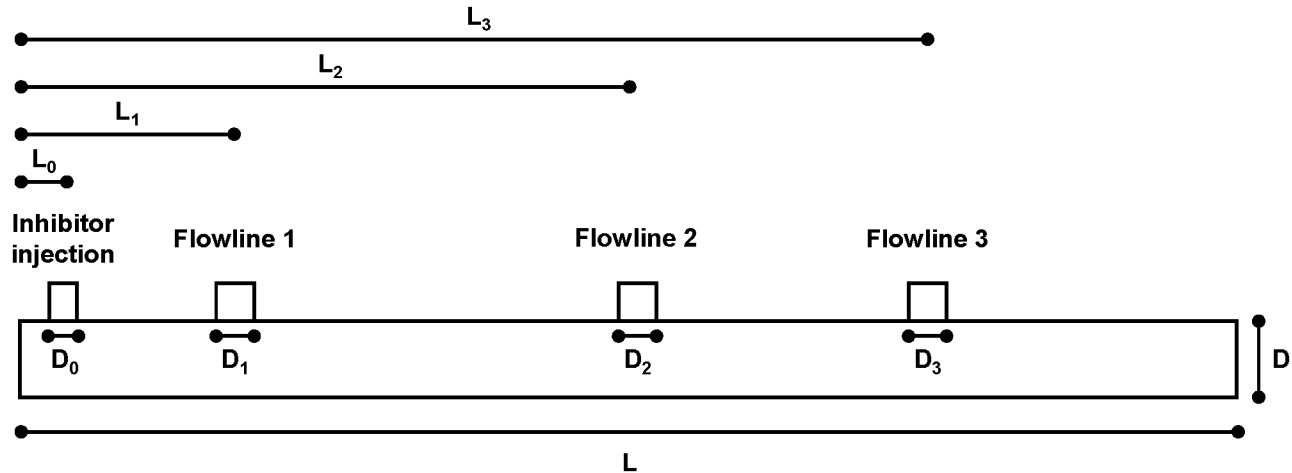


Figure 1—Schematic diagram of a trunkline system with three flowlines

We collapse the 3-dimensional pipe system into a simplified 2-dimensional channel in the x - z plane. Figure 2 shows the numerical implementation of the trunkline system. The numerical implementation considers the trunkline inlet, inhibitor line, and each of the three flowlines as line sources along the upper boundary, with length equal to their corresponding pipe ID (Fig. 2, see relevant dimensions in Table 1). We mesh the model with COMSOL's flow-controlled "coarser" scheme and locally refine around the main inlet and all line sources to balance stability, accuracy, and runtime. The model injects both scale inhibitor and scaling ions at specified concentrations, with scale inhibitor only being injected from the inhibitor line and scaling ions injected from all three flow lines. The numerical model assumes isothermal conditions (i.e., constant temperature), with temperature equal to $T=60^{\circ}\text{C}$. At those specific conditions, the density and viscosity of water and oil are $\rho_{\text{water}}=1000 \text{ kg/m}^3$, $\mu_{\text{water}}=0.5 \text{ mPa}\cdot\text{s}$, and $\rho_{\text{oil}}=800 \text{ kg/m}^3$, $\mu_{\text{oil}}=1.5 \text{ mPa}\cdot\text{s}$.

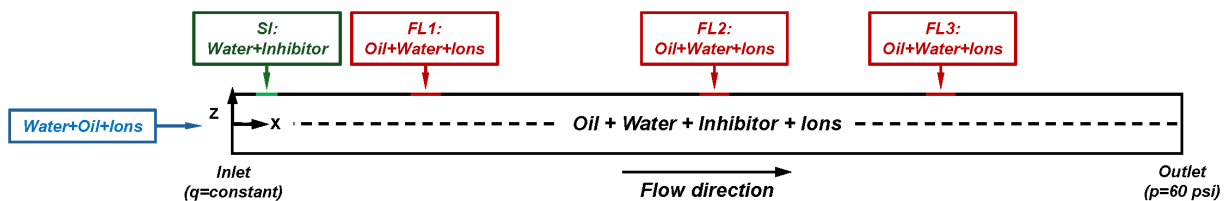


Figure 2—Physical model of an oil-water trunkline system with three flowlines.

Note: SI = scale inhibitor, FL = flow line, q = flowrate, and p = pressure

Initial and Boundary Conditions. At the initial condition, the model assumes zero velocity at the trunkline inlet and all three flowlines; the channel initially contains stagnant oil and zero inhibitor and scaling ion concentration.

Table 2 highlights the flow properties for each boundary in the simulation. At the upstream end of the trunkline, we impose an inflow boundary whose total volumetric rate and water-cut match the values listed in Table 2. The model also considers each of the three production flowlines as inflow boundaries, with individual flow rates and water-oil fractions. Additionally, each flowline injects two scaling ions with specified concentrations. A separate inhibitor-injection line discharges 50 L/d of water containing 10,000 ppm of scale inhibitor and is therefore defined as an additional inflow boundary along the top wall of the channel. At the end of the trunkline, we define an outlet pressure boundary fixed at $p=60 \text{ psi}$. Lastly, the model considers a no-slip boundary condition at the trunkline walls (note: the inhibitor and ions are introduced exclusively within the water phase).

Table 2—Trunkline system flow properties

Property	Trunkline inlet	Flowline 1	Flowline 2	Flowline 3
Flow rate of oil-water mixture [bbl/d]	35,000	3,000	7,000	2,500
Water cut [%]	20	10	55	45
Concentration of scaling ion 1, C1 [ppm]	1,000	1,000	2,000	3,000
Concentration of scaling ion 2, C2 [ppm]	2,000	1,500	3,000	2,500

Results and Discussion

Inhibitor Concentration Transient (Base Case)

Figure 3 shows how the inhibitor concentration evolves with time for the base case (10,000 ppm at 50 L/d). The concentration front translates downstream as time advances, showing a localized high-concentration plume near the injection line that dilutes downstream (Fig. 3a). The inhibitor concentration distribution reveals a stepwise pattern, where each step occurs at a flowline entry due to the addition of an oil-water stream that dilutes the inhibitor in the water phase (Fig. 3b, Note: $z=0.2\text{m}$ for all line graph plots). The sharp gradients in concentration after the flowlines confirm that turbulent mixing governs transport, while molecular diffusion remains negligible, consistent with the modeling assumptions. After the last tie-in, the concentration remains relatively constant towards the outlet. Steady state is reached after approximately 1 h of injection under the specified operating conditions, so we report all remaining results at that time.

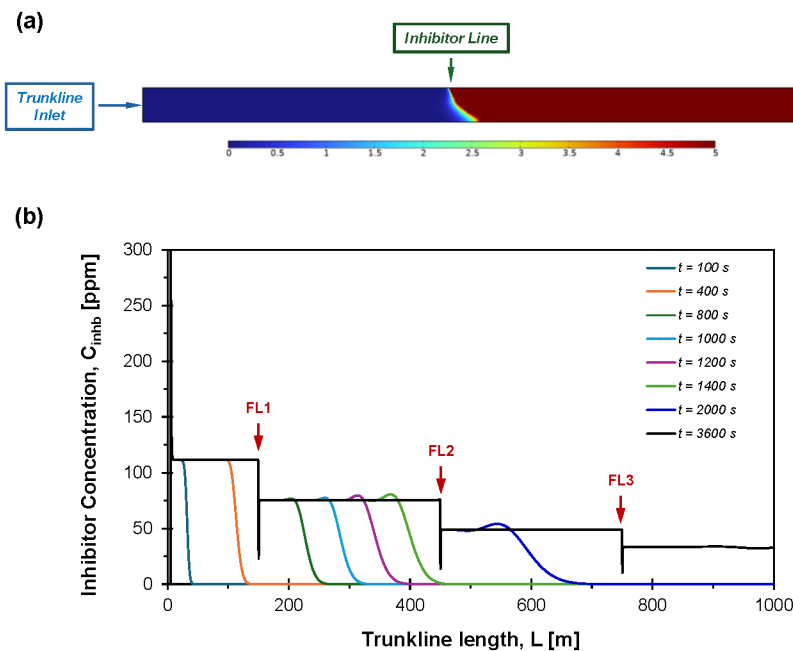


Figure 3—Inhibitor-concentration transient for the base case (10,000 ppm injected at 50 L/d). (a) Inhibitor concentration near the injection point at $t = 3600$ s. (b) Inhibitor concentration profiles at successive times ($t = 100, 400, 800, 1000, 1200, 1400, 2000$, and 3600 s). Stepwise drops mark the three flowlines, where incoming oil-water streams dilute the inhibitor in the water phase.

Steady-State Scaling Ion Distribution and Local Hydrodynamics

Figure 4 highlights the concentration distribution of scaling ions at steady-state under the base case conditions. Data shows that both species exhibit a pronounced wall-attached plume just downstream of the flowline (Fig. 4a). The plume decays axially along the trunkline length; yet, the elevated concentrations signal a high-risk zone for scale nucleation right after the flowline tie-in. The concentration profiles show stepwise jumps that coincide with the flowlines tie-ins due to the input of additional dissolved ions

(highlighted in Table 2). Between tie-ins the curves settle into nearly flat plateaus, indicating that turbulent mixing quickly homogenizes concentrations after the initial dilution spike.

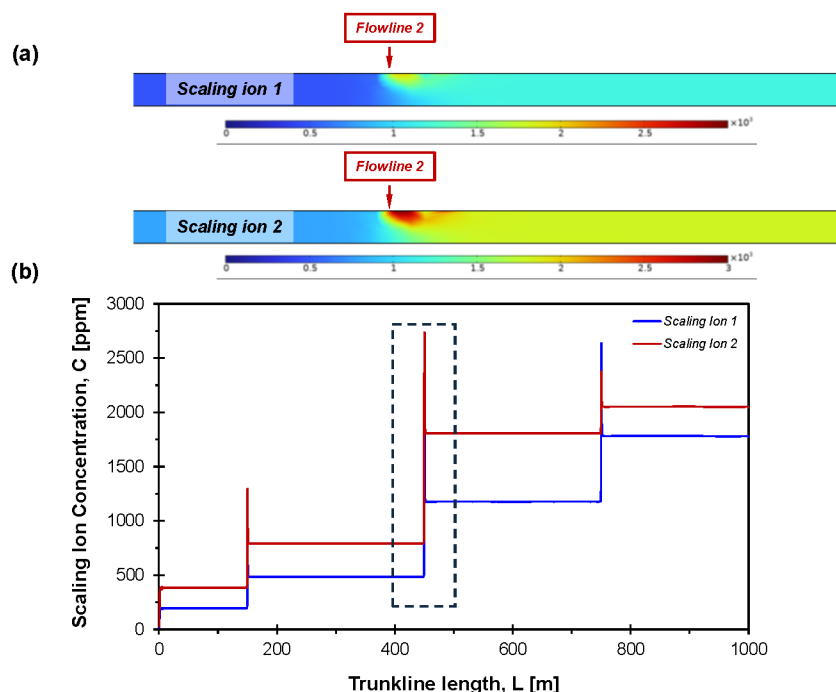


Figure 4—Steady-state scaling ion concentration for the base case (inhibitor 10,000 ppm, 50 L/d). (a) Two-dimensional contours near Flowline 2. (b) Scaling ion concentration profile. Stepwise increase marks the three flowlines, where incoming inputs increase the ion concentration.

To understand why the plume persists even after reaching steady-state, Figure 5 illustrates the mixture flow velocity and water-cut at the same location (near flowline 2). The tie-in produces a recirculating vortex with low axial velocity and elevated water volume fraction (highlighted by the dashed box). Low shear hinders convective removal of ions, while the higher water content supplies more dissolved species. Because all ions reside exclusively at the water phase, this condition greatly increases the likelihood of scale precipitation along the upper wall. Such spatial details would be inaccessible without a multiphase CFD model, as routine point sampling would miss both the magnitude and the precise location of these high-risk zones.

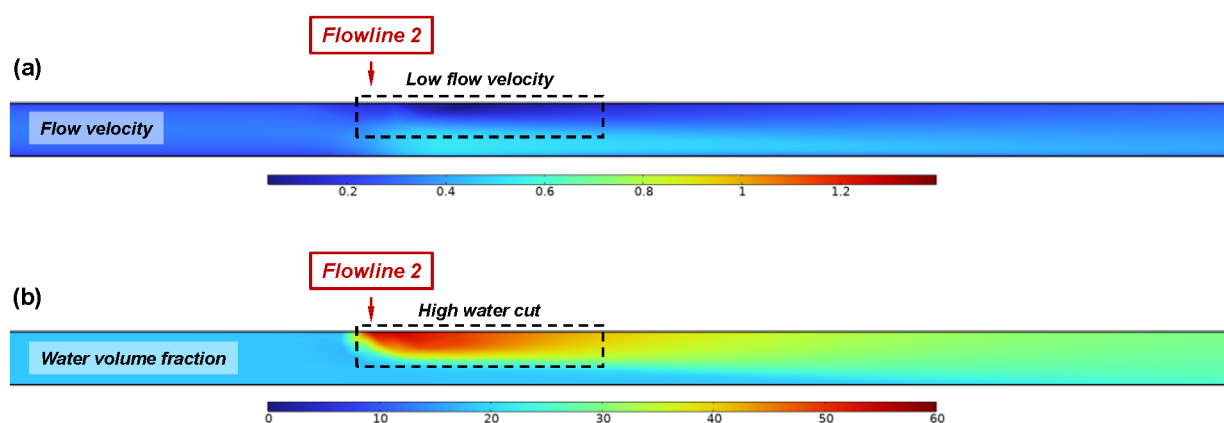


Figure 5—Local hydrodynamics downstream of Flowline 2 at steady state. (a) Flow velocity contours. (b) Water volume fraction contours. Simulations highlight a low-velocity recirculation zone with high-water-cut (dashed box).

Effect of Increasing Injection Concentration on Scale Inhibitor Front

To give field engineers a practical guide, we now test raising the inhibitor concentration while maintaining a constant injection rate ($q=50$ L/d). Figure 6 plots the steady-state concentration profiles for four inlet inhibitor concentrations: 1,000, 5,000, 10,000 (base case), and 50,000 ppm. The response is intuitive yet operationally useful: every doubling of the inlet concentration shifts the entire profile upward by roughly the same factor, so the stepwise jumps caused by the three flowlines remain but sit progressively higher on the y-axis.

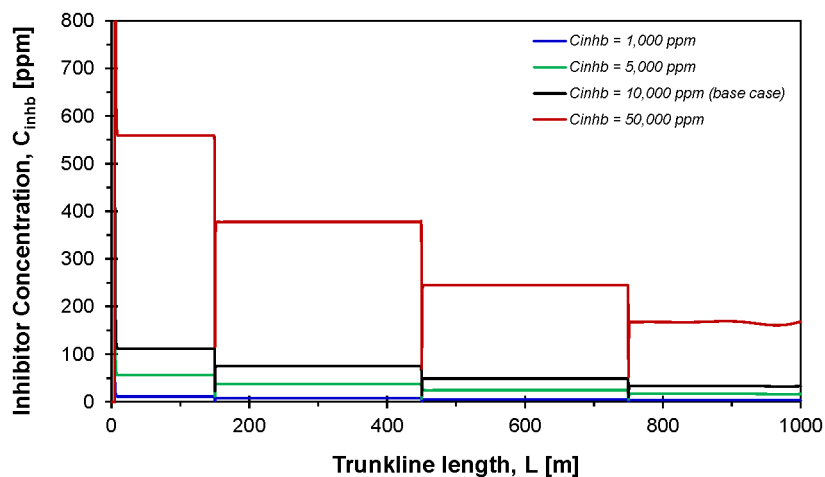


Figure 6—Steady-state inhibitor concentration along the trunkline for four inlet concentrations at a fixed injection rate of 50 L/d.

Comparing this result with our previous publication in which we raised the injection rate instead of the concentration (Liu et al., 2025), we observe a similar qualitative trend: increasing both flowrate and concentration scales the entire profile uniformly. In practice, the optimal set-point pairs the expected scaling-load with the minimum inhibitor residual required to suppress precipitation.

Conclusions

This study extends our earlier work by simulating the concurrent transport of a water-soluble scale inhibitor and scaling ions in a turbulent oil-water trunkline fed by three production flowlines. The coupled CFD model resolves two-phase hydrodynamics, turbulence, and multi-species convection-dispersion, allowing us to visualize effects that routine field monitoring cannot capture. Key findings are as follows:

1. Stepwise concentration profiles originate at every tie-in. Both inhibitor and ion curves exhibit discrete jumps (dilution for inhibitor, enrichment for scaling ions) where the incoming flowlines merge with the trunkline. Between tie-ins, turbulent mixing quickly flattens the profiles, confirming that turbulent mixing dominates over molecular diffusion for species transport.
2. Tie-in hydrodynamics create localized scale-risk hotspots. CFD reveals a low-velocity recirculation zone with elevated water cut immediately downstream of each flowline. Because all ions reside exclusively in the water phase, these zones accumulate dissolved ions while experiencing poor convective flow, favoring scale nucleation along the upper wall.
3. Increasing the inhibitor stock concentration uniformly lifts the entire concentration field. At a fixed injection rate, raising the inlet concentration from 1,000 ppm to 50,000 ppm boosts the outlet residual from ~15 ppm to ~240 ppm while preserving the stepwise pattern. Stock concentration and flow rate therefore provide two independent operational parameters.

4. Optimal dosing must balance inhibitor cost against ion load. The CFD framework enables rapid analyses that match site-specific water chemistry, tie-in geometry, and flow conditions. By iterating stock concentration and injection rate, engineers can meet a minimum residual target while avoiding chemical overspend.
5. Model fidelity hinges on explicit ion tracking. Including individual ions, rather than a lumped salinity term, captures local concentration spikes that drive precipitation thermodynamics. The approach also lays the groundwork for future coupling with geochemical reaction kinetics.

This study demonstrates that the presented model and the CFD-based simulations enable the optimization of the scale inhibitor injection strategy, which matches site-specific water chemistry, tie-in geometry, and flow conditions. The findings provide actionable insight for pipeline-flow-assurance programs. Future work will incorporate inhibitor-ion complexation and precipitation reactions to predict not only where scaling is likely to occur but also the reaction kinetics, thereby closing the loop between transport modelling and scale-growth forecasting.

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